

- trajna, velika deformacija

tekočine: kapljnine (liquid)
(fluid) plini

- stisljivost $\frac{\Delta V}{V} = -\chi \Delta p$

- (velika) za pline

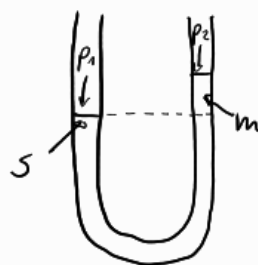
- majhna za kapljnine

teko	snov
m	$\rho = \frac{dm}{dV}$
$\vec{F}$	$p = \frac{F}{S} \left[\frac{N}{m^2} = \frac{J}{m^3} \right]$

	kapljnina	plin
sistem	vezan	prost
W	$W_k + W_p < 0$ $\uparrow$ $> 0; < W_{pl}$	$W_k + W_p > 0$ $\uparrow$ $> 0; < 0$ $\neq W_{pl}$

Pascalov zakon - p je skalar

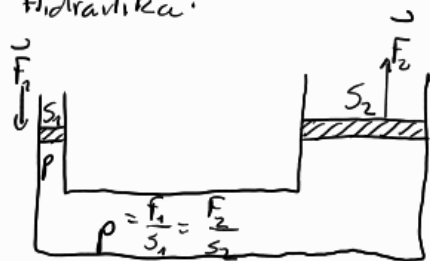
(Tlak v tekočini deluje v vse smeri in na stene)



$$(p_2 - p_1)S = mg = \rho \cdot S \cdot h \cdot g$$

$$\Delta p = \rho g h$$

Hidravlika:



$$F_2 = \frac{S_2}{S_1} F_1$$

- zvezni ojačevalci sile (lahko dobimo poljubno razmerje sil)

- hidravlika, vzvodi

- škripec ni zvezni oj. sile, ker lahko dobimo le razmerja 2, 3, ...

Mirujoča tekočina $\rightarrow$ hidrostatika $\rightarrow$ hidrostatski tlak



$$\Delta p = \rho g h$$

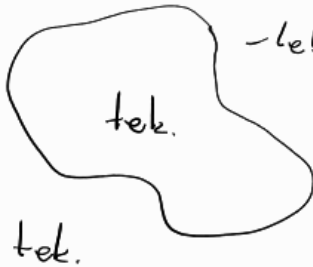
$$p = p_0 + \rho g h$$

Vzgo4 (Arhimed)

$$\vec{F}_{\text{tek}} = \int p d\vec{S} = \vec{F}_v$$



nam. telesa vzamemo
"vrečke s tekočino"



$$\text{lebditi} \Rightarrow \sum \vec{F} = 0$$

$$\vec{F}_g = \vec{F}_{\text{tek.}}$$

$$\rho_{\text{tek.}} g V = \int p d\vec{S} = F_v$$

$$\boxed{F_v = \rho_{\text{tek.}} g V}$$

Površinska napetost

- v vesolju: voda $\rightarrow$ krogla (stabilno ravnovesje) $\rightarrow$ dobajajo haka W_p , ki ima min pri najm. površ.
 $\downarrow$
 min površina

$$W_{p,\text{por}} \propto S \rightarrow \boxed{W_{\text{por}} = \gamma S}$$

koeficient pov. nap.

- v zrak: medmolekulske sile (privlačne) $\rightarrow W_p < 0$

$$\text{oooo} \quad W_{p,\text{not}} = N_{\text{sosed.}} W_{p,1}$$

$$\text{oooo} \quad W_{p,\text{zun.}} \sim \frac{N_{\text{sosed.}}}{2} \cdot W_{p,1}$$

$$W_p = \underbrace{N(W_{p,\text{not}})}_{<0} - \underbrace{N_{\text{pov.}} \frac{W_{p,\text{not}}}{2}}_{>0} \Rightarrow \underline{W_{p,5} = \gamma S}$$

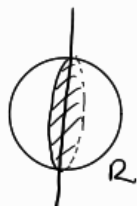
$$A = F dx = dW_s = \gamma dS = \gamma l dx$$

↓

$$F = \gamma l$$

↓
dolžina stika z opno

kapljača



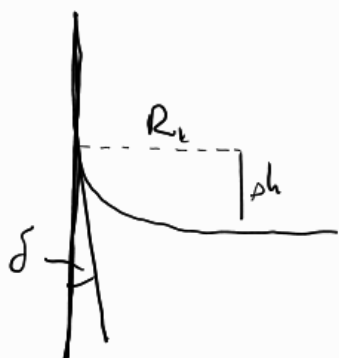
$$F = \Delta p \pi R^2 = \gamma 2\pi R$$

↑
tako jo
vleče
narazen

skupaj jo drži
pov. nap.

$$\Delta p = \frac{2\gamma}{R}$$

Kapilarni dvig:



steklo

	Voda	0
	parafin	107°
	Hg	140°
Ag	Voda	90°

$\sigma > 90^\circ$



$$\Delta p = \frac{2\gamma}{R} = \rho g \Delta h$$

$$\Delta h = \frac{2\gamma \cos \sigma}{\rho g R}$$

Gibanje tekočin

hidrodinamika
gibanje → tok



tokovnice - se ne sekajo!
- če "vzporedne" - grejo od zač. do konca
- če sklenjene - vrtinci

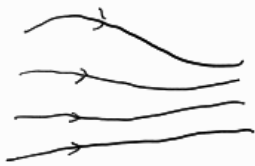
tok
laminarni
turbulentni

če tok odvisen od časa - stacionarni tok, drugače nestacionarni

- turbulenten tok vedno nestac.

- laminarni tok ponavadi stacionaren, v pos. obliščinah lahko nestac.

Opis: (vektorsko) hitrostno polje: $\vec{v}(\vec{r}, t)$



SLIKA PO MERITVI



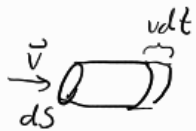
TANGENTA NA TOKOVNICE
(SMER)

velikost vektorjev:

pretok:

$$\Phi_v = \frac{dV}{dt} \quad (\text{prostorninski})$$

$$\Phi_m = \frac{dm}{dt} = \rho \frac{dV}{dt} \quad (\text{masni})$$



$$\frac{d\Phi_v}{dt} = \frac{dV}{dt} = \frac{dS \cdot v dt}{dt} = \vec{v} \cdot d\vec{S}$$

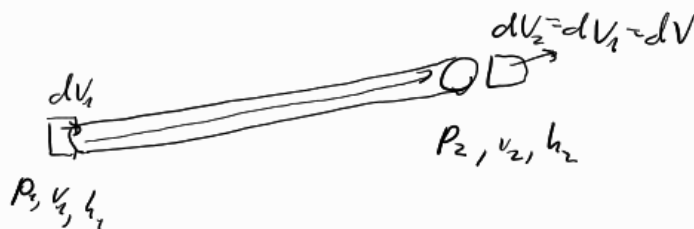
$$\Phi_v = \int \vec{v} \cdot d\vec{S}$$

$$W_k = \int dm \frac{v^2}{2} = \int \rho dV \frac{v^2}{2}$$

- ni disipativnih sil
 - nestisljiva tekočina
 - laminarni stacionarni tok

} voda ✓
 } zrak ✓

gostota kin. en.: $w_k = \frac{dW_k}{dV}$



$$dA = F ds = \frac{dF}{dS} dS ds = \rho dV$$

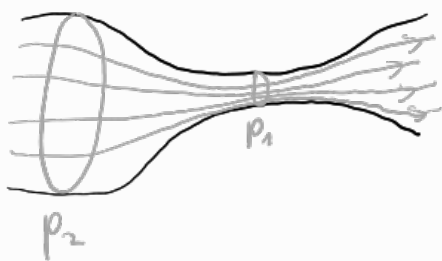
$$A = p_1 dV - p_2 dV = dm \left(\frac{v_2^2}{2} - \frac{v_1^2}{2} \right) + g dm (h_2 - h_1) = \Delta W_k + \Delta W_{p,g}$$

$$p_1 + \rho \frac{v_1^2}{2} + \rho g h_1 = p_2 + \rho \frac{v_2^2}{2} + \rho g h_2$$

$$\Rightarrow \underline{p + w_k + w_p = \text{konst.}}$$

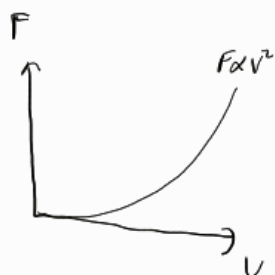
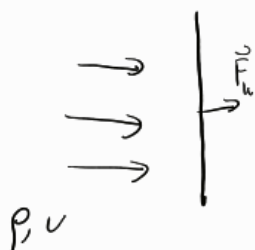
Bernoullijeva enačba

hidrodinamski paradoks:



$$p_2 > p_1$$

Vpor



$$F_u = \Delta p S =$$

$$= (p - p') S = \rho \frac{v^2}{2} S$$

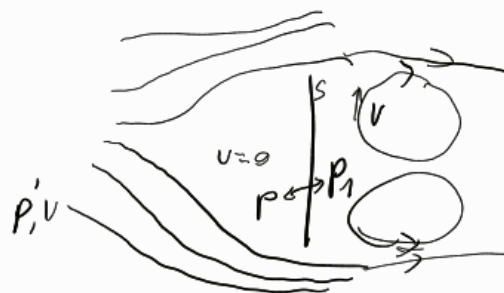


v realnosti:

$$F_u = \rho \frac{v^2}{2} S \cdot c_u$$

Kvadratni zakon upora

koeficient, ki ga dobimo iz meritev



$$p = p' + \rho \frac{v^2}{2}$$

$$\Delta p = p - p_1$$

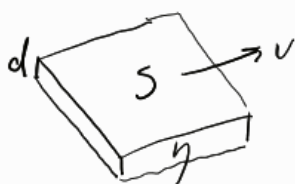
$$p_1 = p'$$

$$\Delta p = p - p' = \rho \frac{v^2}{2}$$

ker se veter v vrtilnem gibanju scca. isto hitrostjo kot daleč od osi

Viskoznost

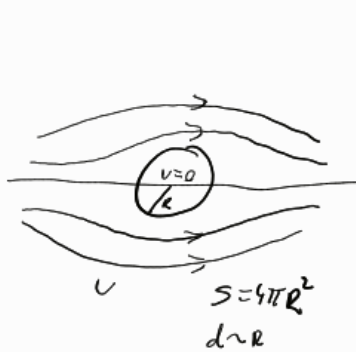
η -koeficient viskoznosti



$$\frac{F}{S} = \eta \frac{v}{d}$$

$$\frac{v(y+dy)}{v(y)} dy$$

$$\frac{F}{S} = \eta \frac{dv_y}{dy}$$



$$F_u = \eta \frac{v}{R} \cdot 4\pi R^2$$

gre v 6 zaradi napake

$$F_u = 6\pi R \eta v$$

Stokesov zakon (za kroglico)

↑
linearni zakon upora

ZA KROGLJO:

$$F_u^{(u)} = \frac{1}{2} c_w \rho v^2 S \xrightarrow{\text{za kroglo}} F_u^{(u)} = 0,2 \rho v^2 S$$

$$F_u^{(u)} = 6\pi R \eta v$$

$Re > 1000$:

uporabimo kvadratni zakon
 $F_u^{(u)}$

Tok

turbulenten

$Re < 0,1$:

uporabimo linearni zakon
 $F_u^{(u)}$

laminaren

Reynoldsovo število

$$Re = \frac{\frac{1}{2} c_w \rho v^2 R^2}{6\pi \eta R v} = \frac{1}{12} \frac{c_w \rho v R^2}{\eta}$$

avira

$$Re = \frac{2r \rho v}{\eta}$$

relativno med obema

tekočina

⇒ med letalka/maketa se obnašata isto, če je Re enako
npr. manjša velikost, večja hitrost

$$\eta \left[\frac{Ns}{m^2} = \frac{kg}{m \cdot s} \right]$$

$T = 20^\circ C$

zraka $1,7 \cdot 10^{-5}$

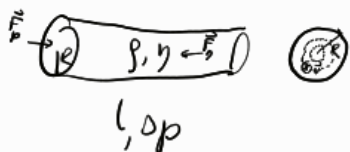
H_2O 10^{-3}

glicerin 0,85

smole $\sim 10^7$

Poiseuille

laminaren stacionaren tok



$$v(R) = 0 \Rightarrow v(r) = ?$$

$$\vec{F}_p = \vec{F}_\eta$$

$$\Delta p S = \Delta p \pi R^2 = \eta \cdot \frac{dv}{dr} \cdot 2\pi r \cdot l$$

$$\frac{dv}{dr} = \frac{\Delta p \pi R^2}{\eta 2\pi r l} = \frac{\Delta p R}{2\eta l} \rightarrow \int_v^0 dv = \frac{\Delta p R}{2\eta l} \int_r^R \frac{1}{r} dr$$

je negativno

$$-v = -\frac{\Delta p R}{4\eta l} (R^2 - r^2)$$

$$v(r) = \frac{\Delta p R}{4\eta l} (R^2 - r^2) \rightarrow v_{max} = v(0) = \frac{\Delta p R^3}{4\eta l}$$

$$v(r) = v_{max} \left(1 - \frac{r^2}{R^2} \right)$$

$$\bar{\Phi}_v = \int \vec{v} \cdot d\vec{S} = v_{max} \int_0^R \left(1 - \frac{r^2}{R^2}\right) 2\pi r dr = v_{max} \cdot 2\pi \left(\frac{R^2}{2} - \frac{R^4}{4R^2}\right) = \boxed{v_{max} \cdot \pi \cdot \frac{R^2}{2}} = \bar{v} \cdot \pi R^2$$

$$\boxed{\bar{v} = \frac{v_{max}}{2}}$$

$$\boxed{\bar{\Phi}_v = \frac{\Delta p}{l} \frac{\pi R^4}{8\eta}}$$

← SAMO ZA LAMINAREN TOK TEKOČINE V CEVI

$$\bar{\Phi}_v \propto R^4 \propto S^2$$